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Cambrian paleomagnetism in the Bateni Ridge: evolution of the Kuznetsk Alatau island arc (*southern Siberia*)

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Abstract

We present paleomagnetic data on the Cambrian volcanosedimentary complexes of the Bateni Ridge, which correspond to the evolution of the Kuznetsk Alatau island arc. The paleomagnetic poles which served as the basis for the apparent polar-wander path (APWP) were determined. The paleomagnetic data were compared with those on the Cambrian poles for other fragments of the Kuznetsk Alatau island arc. For example, the APWP of the Bateni fragment corresponds to that of the Kiya (Martaiga) fragment of the eastern slope of the Kuznetsk Alatau. The distribution of the paleomagnetic poles suggests that the relative drift of the Kuznetsk Alatau arc fragments and the entire island arc system resulted from large-amplitude strike-slips. In the Cambrian, this paleoarc moved from ~10° N to ~10° S and rotated more than 50° clockwise. In the paleomagnetic record, its accretion is reflected in the coincidence of the Late Cambrian–Early Ordovician poles on the eastern slope of the Kuznetsk Alatau with one another and the coeval poles for the Siberian craton. This coincidence also suggests that the general structure of the present Kuznetsk Alatàu formed as early as the Early Caledonian. Nevertheless, the differences in the position of the poles on the western and eastern slopes of the Kuznetsk Alatau suggest that strike-slip activity along the Kuznetsk–Altai deep fault (the major one in the region) continued in the Late Paleozoic and, probably, in the Mesozoic.
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Keywords: Cambrian; paleomagnetic pole; strike-slip activity; Kuznetsk Alatau paleoisland arc; Siberia

Introduction

The Kuznetsk Alatau is one of the largest island arc fragments among the Early Caledonian structures of the Altai–Sayan region; the Cambrian complexes of Gorny Altai are regarded as their southern extensions. In the east and west, it is overlain by the Late Paleozoic sediments of the Kuznetsk and Minusa basins; in the north, by the Meso-Cenozoic sediments of the West Siberian Plate. The volcanoplutonic and sedimentary complexes making up the region were described in detail by many researchers. A comprehensive review of geological data on the Kuznetsk Alatau was given in (Alabin and Kalinin, 1999). The tectonic structure of the Kuznetsk Alatau was depicted as a group of structural units (terranes), each having its own tectonics but all belonging to the same paleoisland arc system (Alabin and Kalinin, 1999; Kazansky et al., 2003a). Here, the structures of the entire lateral series of the island arc system are combined. The marginal-sea complexes which formed in a back-volcanic arc environment are the most widespread. Hereinafter, for brevity sake, this group of rock assemblages, typical of different parts of the island arc system, will be called the Kuznetsk Alatau island arc. The present structure of the region has been formed mainly by strike-slip activity in the constituent terranes. The most remarkable fault is the Kuznetsk–Altai one. It is usually regarded as a roughly N–S trending extension fault, but this fault system at different stages of tectonic evolution operated in different kinematic modes, including strike-slip activity (Alabin and Kalinin, 1999; Berzin, 1995). The deep-mantle origin of the fault permits identifying it with the structure confined to the paleosubduction seismic focal zone (Alabin and Kalinin, 1999). The strike-slip activity is presumed to be most distinct during the accretion of the arc to the craton (Berzin, 1995; Kazansky, 2002). Its influence is also observed later, after the general structure of the Early Caledonian crust had formed in the southwestern Siberian craton (Buslov et al., 2003; Kazansky, 2002; Metelkin et al., 2010; Vladimirov et al., 2003). The Kuznetsk–Altai roughly N–S-trending system

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is associated with roughly E–W-trending lateral strike-slip faults. Their age and formation mechanism are uncertain. One of the explanations is a model presupposing the leading role of a roughly N–S-trending sinistral strike-slip fault under progressive shortening, which led to the breakup of the island arc, the roughly E–W-trending strike-slip activity of isolated fragments, and the final formation of a "horsetail" structure (Kazansky, 2002). Testing these ideas by the paleomagnetic method is one of the main objectives of this study.

Among the major tectonic units in the present structure of the Kuznetsk Alatau, the best studied ones in terms of paleomagnetism are the Zolotoi Kitat (Kazansky, 2002; Metelkin et al., 2000), Kiya (Martaiga) (Kazansky et al., 2003a), and Tersa (Kazansky et al., 2003b) terranes, forming the northern flank of the Kuznetsk Alatau. The more southerly Mrassu and Bateni terranes are not so well-studied. For the Bateni Ridge (the largest fragment of the Kuznetsk Alatau island arc), only individual determinations were obtained (Merkulov, 1982; Metelkin et al., 2000). Here, we present paleomagnetic data on the rocks from seven local stratigraphic units of the Bateni Ridge, which make up the bulk of the Cambrian section.

A brief geological summary of the study area

The Bateni terrane occupies the southeastern Kuznetsk Alatau and comprises Cambrian back-volcanic-arc and back-arc-basin complexes (Kheraskova and Gavrilova, 1996; Koz'min, 2000). Here, typical basalt–andesite–rhyolitic series coexist with abundant thin-bedded cherty-argillaceous-carbonate sediments, shallow-water phosphate-bearing calcarenites, organoclastic and archeocyath-algal limestones, turbidites (predominantly carbonate ones), and various sea slide assemblages with tuff interbeds. Also, Late Precambrian complexes corresponding to intraoceanic rises are observed (Koz'min, 2000) along with the Late Cambrian–Ordovician granitoid plutons sealing the structure of the region (Rudnev et al., 2004; Vladimirov et al., 2001). The 475 ± 5 Ma granite emplacement peak (Rudnev et al., 2004) marks the formation of the Early Caledonian crust during the accretion of the arc to the Siberian craton. The major deformations in the studied island arc complex of the Bateni Ridge belong to this period.

Paleomagnetic studies were conducted in the basins of the Belyi Iyus, Son, and Verkhnyaya Erba Rivers, in the vast territory of the Mrassu–Bateni (Alabin and Kalinin, 1999) structural-assemblage zone, which included (Koz'min, 2000) three large macroblocks (subzones): Kommunar, Iyus–Bateni, and Bateni–Azyrtal (Figs. 1, 2).

Within the Kommunar subzone, the most widespread Cambrian complexes include the Shipilinsk unit, sampled on the left side of the Izekiyula River, on the slope of the Znamenitovskie Gol'tsy Ridge (Shipilinsk locality 55.41° N 89.19° E; 54.44° N 89.23° E). The Shipilinsk unit consists of basalt–andesite–rhyolitic volcanics, tuffs and tuffaceous-terrigenous sediments, associated with the gabbro-diorites of the comagmatic Byuya complex. The entire assemblage is a single homodromous volcanic cycle, which formed owing to a gradual change from submarine (predominantly mafic) to subaerial volcanism, represented by a diverse group of rocks. Scarce petrological and geochemical data suggest the island arc origin of the volcanics and their relationship with the earliest stage in the development of the Kuznetsk Alatau island arc (Koz'min, 2000). Finds of brachiopods, trilobites, and sponge spicules indicate an age no older than the Atdabanian (Koz'min, 2000). The upper age limit is uncertain. According to the Legend for the Minusa Series of Sheets for the Geological Map of Russia, Scale 1 : 200,000 (Makhlaev and Bezzubtsev, 1997), the Shipilinsk unit is located within the Sanashtygol–Agata Horizons, between the Lower and Middle Cambrian. Thus, according to the modern scale (Gradstein et al., 2004), the sediments are no younger than ~510 Ma, and the studied interval correlates with the Botomian.

The age counterpart of the sampled interval of the Shipilinsk unit within the Iyus–Bateni subzone is the Kolodzhul Formation (Fig. 2). It consists of volcanic-terrigenous-carbonate sediments, which accumulated under shallow-water conditions with a strongly dissected topography (Koz'min, 2000). The latter might have resulted from sporadic volcanism. The volcanic part of the section is dominated by amygdaloidal basalts, marked by high Ti contents, high Na alkalinity, and moderate alumina contents, as is typical of back-arc and back-arc basin environments. These basalts were sampled on the right side of the Belyi Iyus River, south of Efremkino Village (Efremkino locality 54.45° N 89.48° E), where one of the representative sections of the formation is located. The stratigraphic position of the unit, corresponding to the Sanashtygol Horizon (Fig. 2), was determined from massive trilobite finds as well as archeocyaths, hyoliths, and brachiopods (Koz'min, 2000). The Kolodzhul Formation is overlain, often with an angular unconformity, by a similar volcanosedimentary complex belonging to the Efremkino Formation (Fig. 2). The latter is marked by the presence of trachyandesites in the section.

The Middle Cambrian Bezymyannaya Formation at the Efremkino locality was sampled 3 km east of the Kolodzhul sediments (54.46° N 89.51° E). The section consists of rhythmically bedded sandstones and siltstones, predominantly volcanomictic ones, rarely gravelstones, and cherts with thin limestone beds, gravitating toward the upper parts of the rhythms. The contact with the underlying Efremkino Formation is conformable (Koz'min, 2000). The flyschoid and turbidite sediments of the Sladkie Koren'ya and Tolcheya Formations are presumed to be the age and facies equivalents of the Bezymyannaya Formation (Fig. 2). Lithofacies studies suggest that the Middle Cambrian turbidites formed in a moderately deep sea basin, in a back-arc environment (Kheraskova and Gavrilova, 1996; Koz'min, 2000). The upper age limit of the Bezymyannaya Formation is uncertain, but the lithologic features and trilobite fauna constrain it within the Mundybash Horizon (Fig. 2).

Within the Bateni–Azyrtal subzone, we studied the upper part of the Cambrian section of the Bateni Ridge. On the left side of the Erba River, 2 km north of Verkhnyaya Erba Village (Fig. 1), the flyschoid and volcanomictic terrigenous-carbonate sediments of the Sladkie Koren'ya Formation are exposed

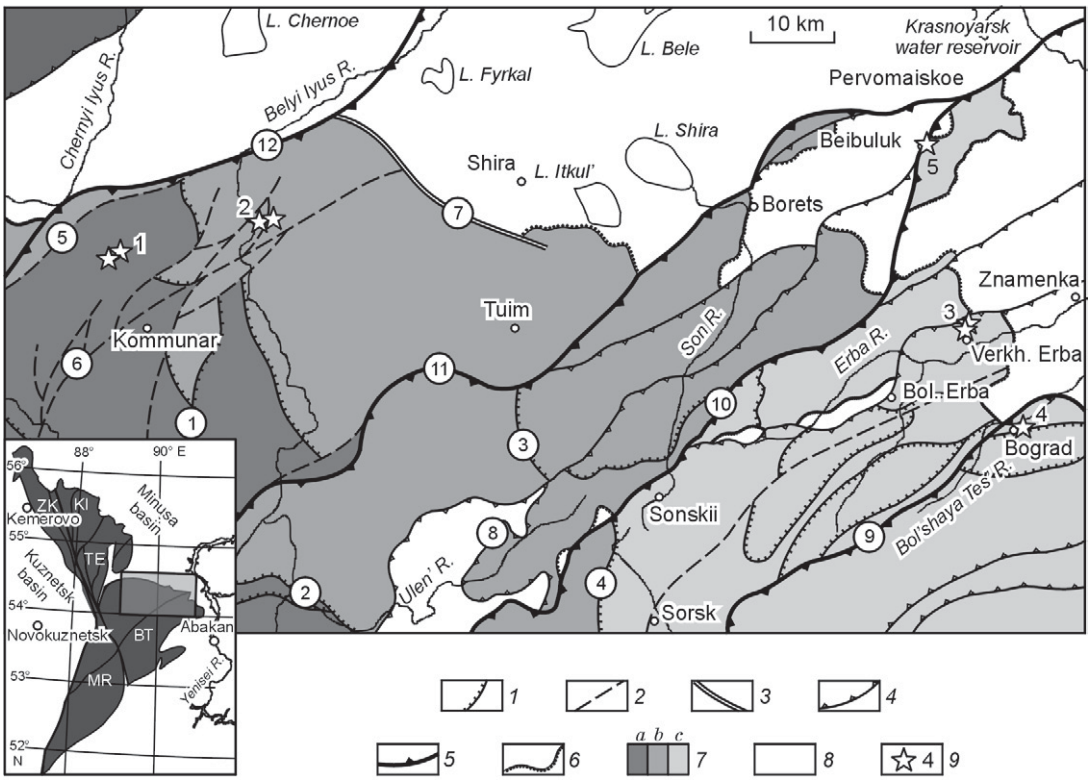


Fig. 1. Tectonic zoning of the central Bateni Ridge, simplified after (Koz'min, 2000). Inset shows the geographic location of the study area and the boundaries of the Kuznetsk Alatau terranes. *1*, Late Cambrian–Early Ordovician thrusts (1, Syya; 2, Belyi Iyus; 3, Son; 4, Sora); *2*, Middle–Late Ordovician strike-slip faults (5, Shipilinsk–Sisim; 6, Kommunar); *3*, initiation faults of the Early Devonian basins (7, Koshkulak–Shira); *4*, post-Early Carboniferous reverse faults–thrusts (8, Ulen'); *5*, long-lived faults of complex kinematics, predominantly thrusts and reverse faults–thrusts (9, Azyrtal; 10, Ulen'–Bateni; 11, Tyukhterek; 12, Chernyi Iyus); *6*, unconformity boundaries; *7*, Late Precambrian–Early Paleozoic compositional complexes of the lower structural stage: *a*, Kommunar subzone; *b*, Iyus–Bateni subzone; *c*, Bateni–Azyrtal subzone; *8*, Late Paleozoic–Mesozoic compositional complexes of the upper structural stage; *9*, paleomagnetic-sampling sites: 1, Shipilinsk; 2, Efremkino; 3, Erba; 4, Bograd; 5, Beibuluk. Inset: terranes: BT, Bateni; KI, Kiya; TE, Tersa; MR, Mrassu; ZK, Zolotoi Kitat; rectangle shows the position of the study area, presented on the sketch map of tectonic zoning.

(Erba locality 54.34° N 90.73° E). In the study area, the predominantly terrigenous strata include variegated tuffs or, less often, effusive rocks, mainly felsic ones. The age of the sediments is constrained by the Mundybash trilobite complex (Resolutions..., 1983). On the slopes of Mt. Krasnyi Kamen' (54.37° N 90.76° E), 3.5 km to the northeast, terrigenous, including red-colored, sediments are combined into the Bateni Formation. The studied locality is stratotypic (Yaroshevich, 1962). Here, a thick unit of cherry-red coarse- to medium-grained cross-bedded sandstones is observed, with beds of small-pebble conglomerates and gray sandy limestones. Higher upsection they alternate monotonously and rhythmically with variegated medium- to fine-grained parallel-bedded sandstones and siltstones. The lower boundary with the Tolcheya Formation is conformable, and the upper one is uncertain. Scarce faunal finds suggest the Amgan–Mayan age of the unit (Makhlaev and Bezzubtsev, 1997). Considering this and the position of the sampled horizons in the general section, the rocks might be no older than ~505 Ma.

Middle–Late(?) Cambrian turbidite complexes were previously studied at two localities (Metelkin et al., 2000); they are difficult to divide, structurally similar to the Tolcheya sediments, and located in the basins of the Bol'shaya Tes' and Karasug Rivers. At the Bograd locality, along a dry riverbed, 1 km east of Bograd Village (54.25° N 90.85° E), a section consisting of conglomerates as well as snuff-colored-green siltstones and tuffaceous siltstones with interbeds of sandstones, tuffaceous sandstones, and intermediate-felsic tuffs (Metelkin et al., 2000) correlates with the upper Tolcheya Formation (Fig. 2). This correlation is based on the general geologic structure of the section, the dominance of volcanomictic turbidites and mixed tuff beds inconsistent along strike, which are typical of the Tolcheya Formation (Koz'min, 2000). At the Beibuluk locality (54.57° N 90.72° E), in the riverbed of a dry tributary of the Karasug River, 700 m east of Beibuluk Village (Fig. 1), a thick turbidite–olistostrome unit, dominated by terrigenous rocks and felsic or, less often, intermediate tuffs (Metelkin et al., 2000), is also structurally similar to the Tolcheya Formation. The age of the latter is constrained by the Mundybash–El'dakh complex of organic remnants at the Late Amgan (Resolutions..., 1983). We think that the previously studied sediments at the Bograd and Beibuluk localities might also be late Amgan.

Methods and results of paleomagnetic studies

Laboratory experiments were performed on the equipment of the Paleomagnetic Center (Trofimuk Institute of Petroleum Geology and Geophysics). The equipment included new-gen-

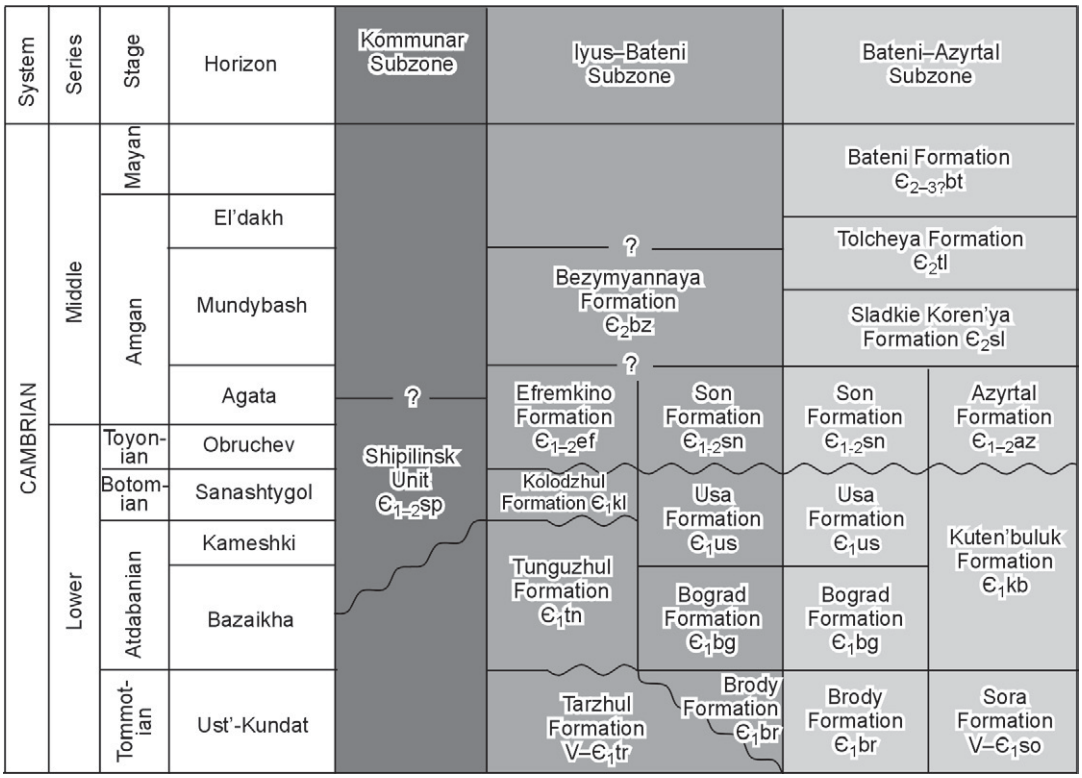


Fig. 2. Fragment of the regional stratigraphic chart (Koz'min, 2000; Makhlaev and Bezzubtsev, 1997).

eration measuring instruments: 2G Enterprise (United States) and HSM (Germany) cryogenic magnetometers along with well-known JR-4 and JR-6 spinner magnetometers (Czech Republic). The experimental technique was standard (Butler, 1992) and based on the stepwise thermal demagnetization (*T*-demagnetization) of all the samples until their complete demagnetization in a shielded furnace. The experimental data were processed by special software, which used common techniques to isolate the components of the remanent magnetization (Kirshvink, 1980; McFadden and McElhinny, 1988); standard statistical techniques for analyzing vectors on a sphere to analyze the paleomagnetic directions (Butler, 1992); different modifications of the fold test (Enkin, 2003; Watson and Enkin, 1993) and the reversal test (McFadden and McElhinny, 1990) to date the remanent magnetization.

The basalts and tuffs of the Shipilinsk locality have a low-intensity natural remanent magnetization (NRM = 0.01–10 mA/m; not typical of volcanics) with multiple components. The orthogonal diagrams are often very difficult to interpret; the unblocking-temperature intervals overlap considerably and have components with different directions. As a rule, complete demagnetization is achieved at ~580 °C (this corresponds to magnetite unblocking temperatures). This mineral might be the main carrier of magnetization. Nevertheless, NRM decay plots often show a typical abrupt decrease in the NRM within a relatively low-temperature interval (~350–400 °C). Sometimes more than half the total magnetization disappears on heating to no more than 300 °C. However, most of the stereoplots show a distinct vector shift along the great circle, especially in the highest temperature part of the demagnetization spectra (Fig. 3). The great circles plotted on the basis of data on sample demagnetization within each sampling site overlap in the same area, suggesting that the NRM has a regular component (Table 1).

Basalt samples from the Kolodzhul Formation, studied at the Efremkino locality and considered equivalents of the Shipilinsk ones, show a much more stable behavior of the NRM vector during the demagnetization. First, they are more magnetized. The NRM intensity never decreases below 10 mA/m; more often it is 100–1000 mA/m, as is typical of igneous rocks. A low-temperature magnetic phase is subtle or absent. Complete demagnetization is achieved at magnetite (~580 °C) and hematite (~690 °C) unblocking temperatures. Note that the directions of the magnetite and hematite components, as a rule, coincide and approach the origin of coordinates on the orthogonal diagram (Fig. 4). Regardless of the unblocking temperature, the characteristic remanent magnetization (ChRM) is inclined negatively, and the site mean direction for stratigraphic coordinates corresponds, within the confidence limits, to that of the regular magnetization of the basalts and tuffs from the Shipilinsk unit (Fig. 4, Table 1).

The mean direction of component A1, established in the Early Cambrian rocks of the Kommunar and Iyus–Bateni subzones, coincides with the previous data (Merkulov, 1982). On the basis of studies conducted within the Efremkino locality, V.P. Merkulov compiled a magnetostratigraphic section, in which the Kolodzhul Formation had an exclusively reverse polarity and the mean direction of the vector (D_s = 3,

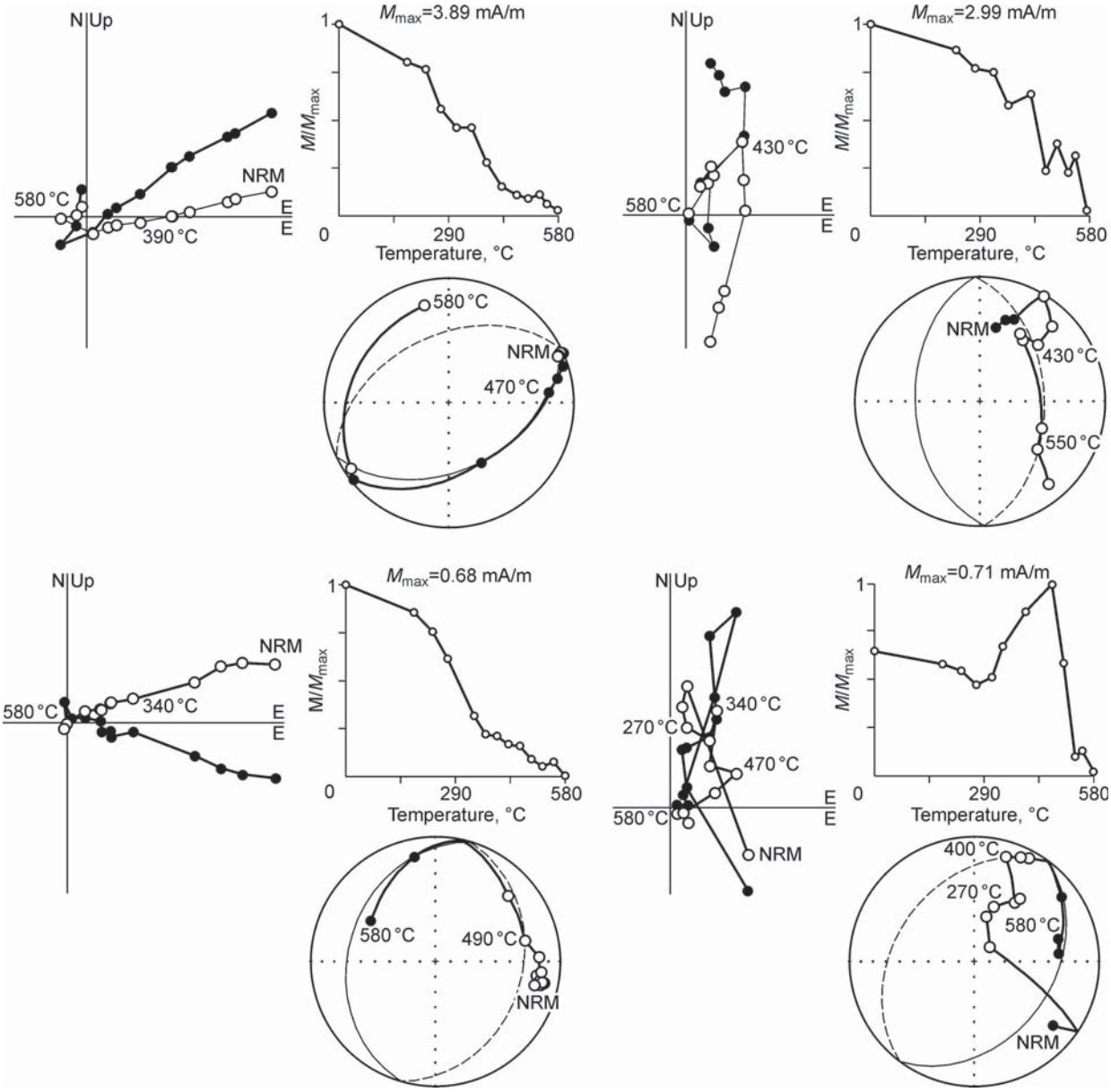


Fig. 3. Typical orthogonal diagrams, associated NRM(*T*) decay plots, and stereoplots, illustrating stepwise thermal demagnetization of the basalts and tuffs from the Shipilinsk unit. Hereinafter, on the orthogonal diagrams, closed (open) circles represent vector endpoints projected onto the horizontal (vertical) plane. In the stereoplots, closed (open) symbols are projected onto the lower (upper) hemisphere; the coordinate system is stratigraphic.

I_s = –11, K = 21.7, α_{95} = 3.1) for 94 samples almost did not differ from that of component A1. Similar paleomagnetic directions were determined for the overlying Efremkino Formation (D_s = 7, I_s = –20, K = 24.7, α_{95} = 3.5) (Merkulov, 1982). The main drawback was the incomplete *T*-demagnetization of the samples, only to 200 °C (Merkulov, 1982). However, as shown by the present studies, the ChRM of the basalts from the Kolodzhul Formation is quite stable, and its direction can be determined immediately after the removal of viscous magnetization; this conditioned the previous positive paleomagnetic result.

The primary origin of component A1 is evidenced by an increase in the precision parameter for the site means directions of magnetization in stratigraphic coordinates (Table 1). Although K_s/K_g = 2.07 < F_c = 2.33 (McElhinny, 1964), the direction-correction fold test (Enkin, 2003) yields a positive result (DC Slope: 1.5236 ± 0.6698). Other types of the fold test are inefficient and do not yield a definite result, because the precision parameter is low and there are slight differences in the bedding of the studied strata. Nevertheless, the above evidence permits interpreting component A1 as Early Cambrian and correlating it with the age of the rocks (~520 Ma).

The sandstones of the Bezymyannaya Formation have relatively high NRM for terrigenous rocks (10^{-3}–10^{-2} mA/m). According to the result of stepwise *T*-demagnetization, except the low-temperature component, removed after heating up to ~300 °C, the ChRM is clearly identified in a relatively

Table 1. Directions of stable remanence magnetization from the Cambrian sections of the Bateni Ridge

No.	Rock, formation, sampling site	$n(nR)/N$	M	D_g	I_g	D_s	I_s	K	α_{95}
Component A1, Iyus–Bateni subzone									
1	Basalts, Kolodzhul, 00bt14	9(9)/10	PCA	10.4	21.4	13.3	–6.4	94.5	5.3
2	Basalts, Kolodzhul, 00bt15	12(12)/14	PCA	33.3	20.5	34.4	–10.2	20.2	9.9
Component A1, Kommunar subzone									
3	Tuffs, Shipilinsk, 00bt2g	6/7	GC	357.7	–8.2	358.7	–14.0	36.1	11.3
4	Tuffs, Shipilinsk, 00bt2m	6/7	GC	20.1	–5.4	20.9	–8.7	114.0	6.3
5	Basalts, Shipilinsk, 00bt3	5/6	GC	357.0	–10.2	357.2	–7.2	110.8	7.3
6	Tuffs, Shipilinsk, 00bt5	4/7	GC	0.4	–23.1	359.1	–27.4	69.5	11.1
7	Basalts, Shipilinsk, 00bt6	7/8	GC	11.8	–6.9	12.9	–15.3	39.7	9.7
8	Basalts, Shipilinsk, 00bt7	4/6	GC	10.5	–8.6	11.7	–17.2	352.4	4.9
9	Basalts, Shipilinsk, 00bt8	11/11	GC	6.1	–6.0	6.9	–15.0	62.9	5.8
Mean A1, 520 Ma				9.6	–2.1	–	–	19.9	11.8
				–	–	10.7	–13.7	37.1	8.6
Component A2, Iyus–Bateni subzone									
10	Sandstones, Bezymyannaya, 00bt9	4/5	GC	358.9	–36.9	343.4	16.0	105.2	9.0
11	Sandstones, Bezymyannaya, 00bt10	15(0)/17	PCA	354.2	–7.3	357.5	12.1	44.4	5.8
12	Sandstones, Bezymyannaya, 00bt11	7(0)/8	PCA	7.7	–22.8	355.2	7.2	48.0	8.8
13	Sandstones, Bezymyannaya, 00bt12	5(1)/7	PCA	29.9	–36.3	348.5	10.0	52.1	10.7
14	Sandstones, Bezymyannaya, 00bt13	7(2)/9	PCA	23.4	–29.0	351.8	8.5	82.1	6.7
Component A2, Bateni–Azyrtal subzone									
15	Tuffs, Sladkie Koren'ya, 01bt28	7(7)/10	PCA	163.7	–9.6	172.1	–14.8	25.8	12.1
16	Tuffs, Sladkie Koren'ya, 01bt29	8(8)/10	PCA	158.4	–21.2	174.8	–15.4	54.1	7.6
Mean A2, 505 Ma				1.3	–15.5	–	–	7.9	22.9
				–	–	351.9	12.0	194.9	4.3
Component A3, Bateni–Azyrtal subzone									
17	Siltstones, Bateni, 01bt33a	9(6)/10	PCA	326.4	–5.3	328.0	16.9	43.4	7.9
18	Sandstones, Bateni, 01bt33b	10(2)/14	PCA	322.1	6.4	326.8	22.2	25.3	9.8
19	Sandstones, Bateni, 01bt34	10(3)z/11	PCA	322.4	29.4	325.9	20.0	42.4	7.5
20	Tuffaceous sandstones, undifferentiated Bograd locality, BA–I*	13(1)/19	CA(1)	292.5	47.7	321.6	33.7	15.8	10.8
21	Tuffs, undifferentiated Beibuluk locality, BA–II*	6(2)/11	CA(2)	299.0	–36.7	307.1	18.1	13.9	19.3
22	Tuffs, undifferentiated Beibuluk locality, BA–III*	6(1)/6	PCA	274.8	–17.7	306.7	34.5	20.4	15.2
Mean A3, 500 Ma				307.4	4.0	–	–	5.0	33.2
					–	319.6	24.5	46.3	9.9

Note. Numbers of the analyses correspond to those in Fig. 8; $n(nR)/N$, the number of unit vectors used in the statistics (including those with a negative inclination in parentheses) to the total number of studied samples; M, analytical methods: PCA, principal–component analysis; GC, great–circle method; CA, combined analysis (the number of circles used in the analysis is in parentheses); D, declination (deg); I, inclination (deg); the indices "g" and "s" show the geographic and stratigraphic coordinates, respectively; K, precision parameter; α_{95}, radius of the cone of 95% confidence.

* After (Metelkin et al., 2000).

high-temperature part of demagnetization spectra. Its unblocking temperatures vary from 540 to 590 °C (Fig. 5). The low-temperature component (D_s = 35, I_s = 3, K = 29.2, α_{95} = 17.3) corresponds to the paleomagnetic direction obtained previously (D_s = 34, I_s = –4, K = 14.2, α_{95} = 4.9), according to *T*-demagnetization up to 200 °C (Merkulov, 1982). This component is not ChRM, and the fold test indicates its postfolding age and, correspondingly, metachronous origin. The high-temperature ChRM is directed differently from the low-temperature component by ~40° in the declination and has a steeper positive inclination (Table 1). For the ChRM distribution, the K_s/K_g ratio is significantly higher (7.95) than the critical one (6.03) within a 99% confidence limit for n = 5 after (McElhinny, 1964). According to the fold test simulation by the (Watson and Enkin, 1993) method, the maximum precision parameter corresponds to 104.2% untilting within a

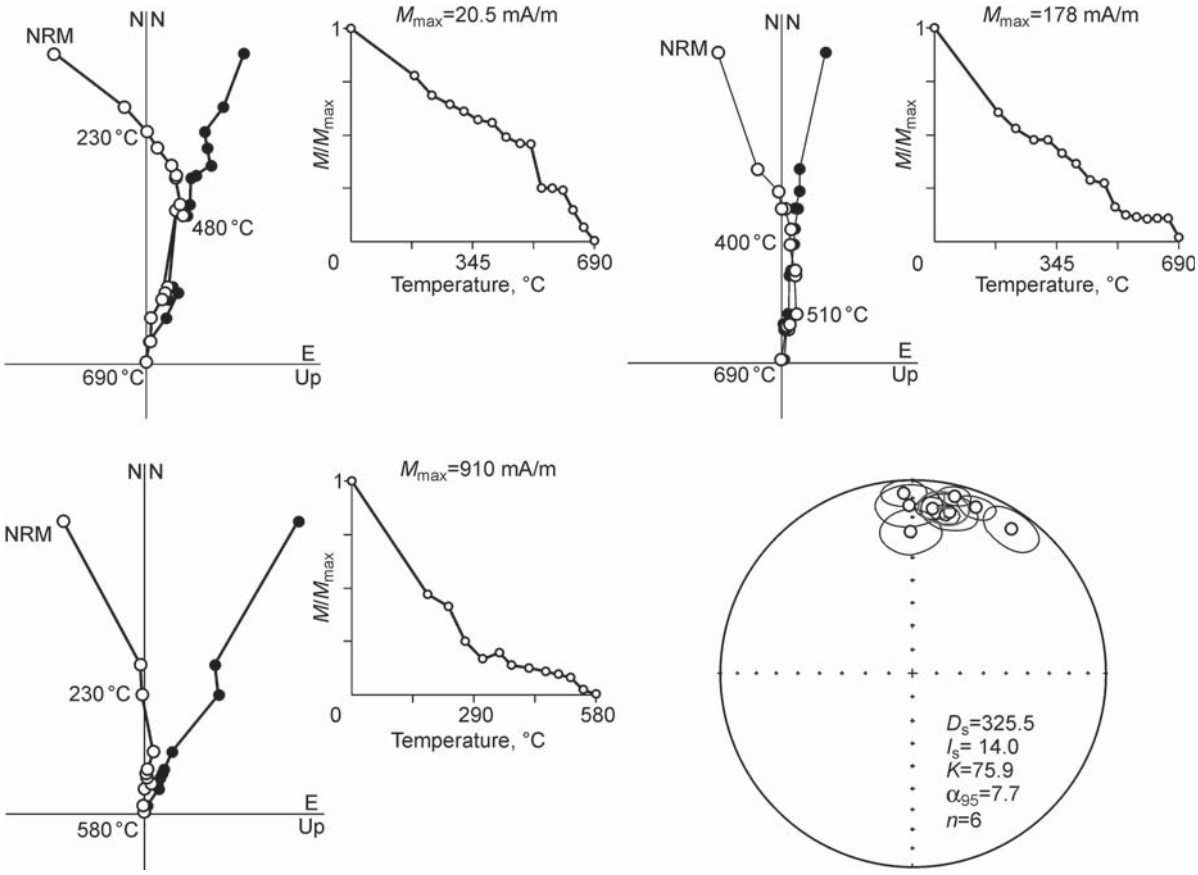


Fig. 4. Typical orthogonal diagrams and associated NRM(*T*) decay plots, illustrating stepwise thermal demagnetization of the basalts from the Kolodzhul Formation; stereoplots show the site mean directions of component A1 from the Shipilinsk unit and Kolodzhul Formation.

narrow confidence limit of 90.8–117.2% (Fig. 5). The direction-correction fold test (Enkin, 2003) also yields a positive result (DC Slope: 1.1596 ± 0.5662), implying that the remanence is probably pretilting.

The tuffs of the Sladkie Koren'ya Formation, coeval with the sandstones of the Bezymyannaya Formation and located at the Erba locality of the Bateni–Azyrtal subzone (Fig. 1), have similar rock-magnetic and paleomagnetic characteristics. The ChRM is removed after heating up to 540–580 °C, and the low-temperature component is often absent (Fig. 6). The distinctive feature is the exclusively negative ChRM inclination (Fig. 6, Table 1).

Comparison between the unit vectors of normal and reversed polarity for the Bezymyannaya and Sladkie Koren'ya Formations (Fig. 6) shows that the magnetization of the rocks might be close in age, that is to say, related to a magnetic reversal during their formation. Note that the secular variation has been adequately sampled and the reversal test yields a positive result: $\gamma = 4.2$ at $\gamma_c = 6.0$, class "B," according to the classification in (McFadden and McElhinny, 1990). The fold test for the site mean directions of component A2 from the rocks of the Iyus–Bateni and Bateni–Azyrtal subzones (Table 1) also indicates the ancient age of the magnetization. The precision parameter for stratigraphic coordinates is considerably higher than that for geographic coordinates: $K_s/K_g = 24.7 > F_c = 2.69$ (McElhinny, 1964). The fold test simulation by the (Watson and Enkin, 1993) method indicates the optimum untilting at $107.9 \pm 10.9\%$. The direction-correction fold test (Enkin, 2003) also yields a positive result (DC Slope: 1.109 ± 0.071). Thus, the reversal and fold tests constrain the age of component A2 as coeval with the formation of the rocks (~505 Ma).

For the red-colored sandstones and siltstones of the Bateni Formation at the Erba locality, the NRM intensity varies from 1 to 100 mA/m; this obviously reflects variations in the concentration of magnetic material. Almost all the NRM intensity is usually removed after heating up to ~580 °C, and the orthogonal diagrams are represented by a single component (Fig. 7). There are several samples in which almost all the NRM intensity is removed after heating up to the hematite unblocking temperature (T_{ub}), but the component removed at 600–680 °C is not regular. Also, the magnetite remanence ($T_{ub} \approx 580$ °C) does not differ from that for the hematite-free samples (Fig. 7). For three studied outcrops, the stable component A3 has both positive and negative inclinations. When reversed to the same polarity, the site mean directions differ from that of component A2 for stratigraphic coordinates by ~25° in the declination and by >5° in the inclination (Table 1).

The previously studied (Metelkin et al., 2000) beds of tuffs and tuffaceous sandstones at the Bograd and Beibuluk localities (as stated above, we correlate them with the Late Amgan

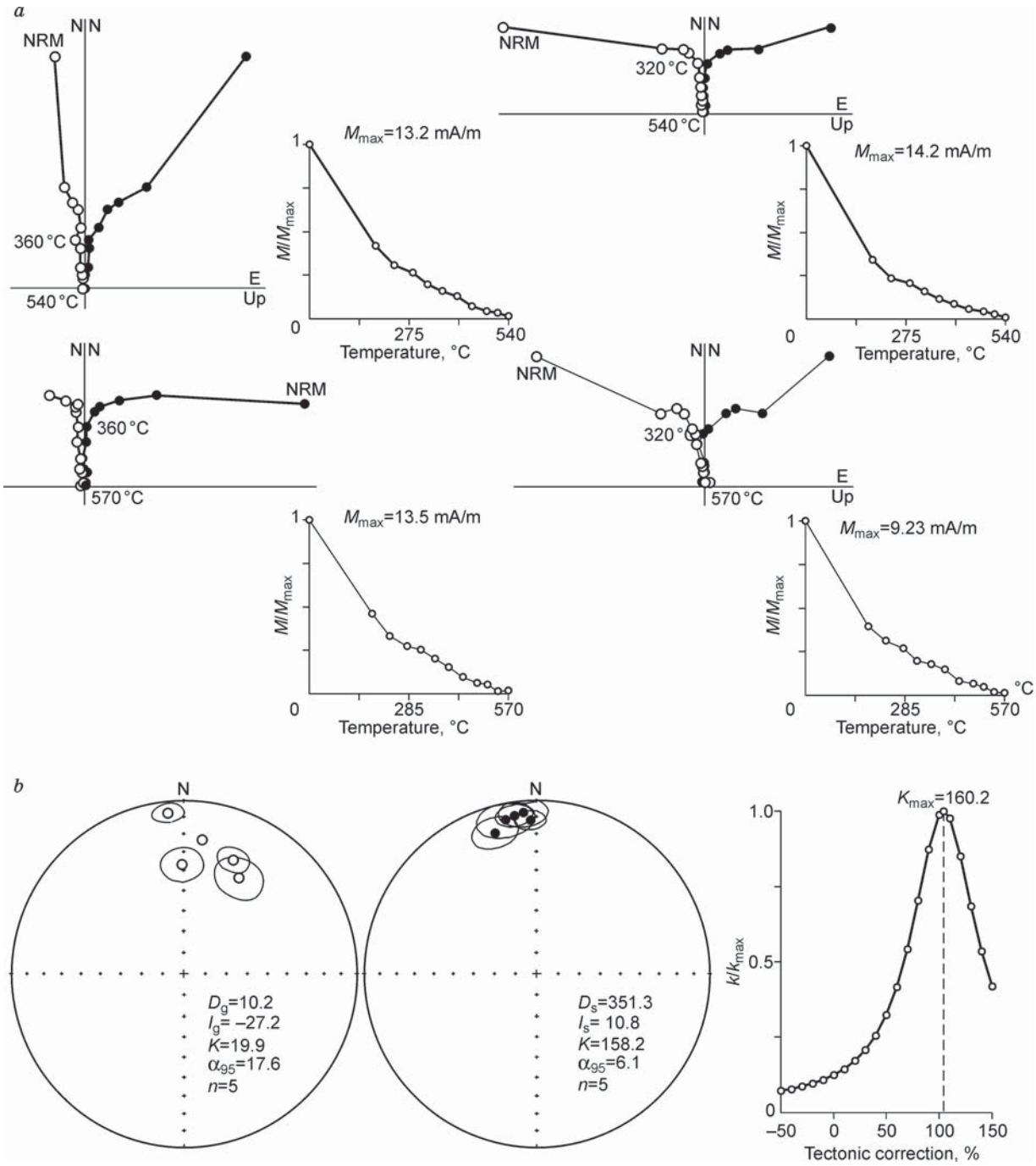


Fig. 5. Paleomagnetic results from the sandstones of the Bezymyannaya Formation: *a*, typical orthogonal diagrams and associated NRM(*T*) decay plots illustrating stepwise thermal demagnetization; *b*, stereoplots shows the site mean directions of component A2 in geographic coordinates (left) and stratigraphic coordinates (right) and a plot showing variations in the precision parameter obtained from untilting by the (Watson and Enkin, 1993) method.

sediments of the Tolcheya Formation, underlying the sandstones and siltstones of the Bateni Formation in the Bateni–Azyrtal subzone) have preserved the component directed almost the same way as component A3 (Table 1). The similarity between the paleomagnetic directions might be indirect evidence confirming our geological observations that these sections correspond to the upper beds of the Tolcheya Formation. Paleomagnetic tests show the synchronous or near-synchronous origin of component A3 for the sandstones of the Bateni Formation and the tuffaceous-terrigenous rocks of the Beibuluk and Bograd localities. The angular distance between the normal and reversed polarity means (Fig. 7) after the reversal is $\gamma = 7.5°$; this is significantly less than the critical value ($\gamma_C = 9.1$). The significance of the result corresponds to class "B," according to (McFadden and McElhinny, 1990). The fold test also yields a positive result (DC Slope: 1.092 ± 0.298 (Enkin, 2003)); the optimum precision is close to 100% untilting ($117.1 \pm 12.9\%$) (Watson and Enkin, 1993).

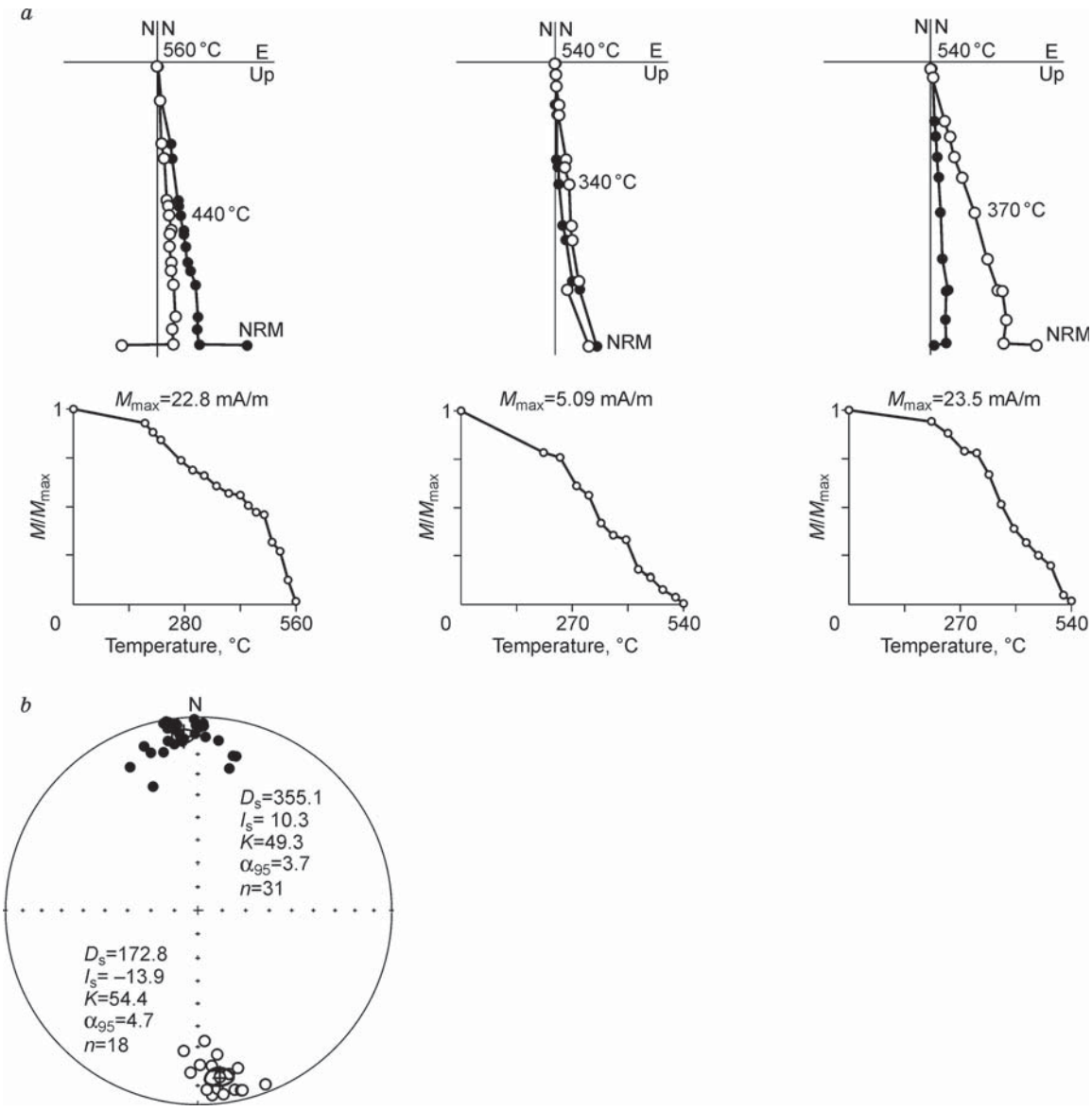


Fig. 6. Typical orthogonal diagrams, associated NRM(*T*) decay plots illustrating stepwise thermal demagnetization of the tuffs from the Sladkie Koren'ya Formation (*a*); stereoplots (*b*) show the distribution of the unit vectors of component A2 from the Sladkie Koren'ya and Bezymyannaya Formations (illustration of the reversal test).

Considering the results of the paleomagnetic tests and the estimated ages of the strata, we presume that component A3 is Amgan–Mayan (~500 Ma).

Discussion of the paleomagnetic data and tectonic implications

Studies of the island arc assemblages in the Bateni Ridge revealed a succession of paleomagnetic directions, which, judging by the above data, reflect the Cambrian island arc tectonics. By the level of detail and the character of apparent drift, the resulting paleomagnetic record can be correlated with the apparent polar-wander path (APWP) for the Kiya terrane (Kazansky et al., 2003a). According to the paleomagnetic data on the Kiya terrane (Kazansky et al., 2003a), the Early Cambrian normal polarity is inclined positively, and the Middle and Late Cambrian ones are inclined negatively (Table 2). In this interpretation, the paleomagnetic directions find themselves on one plane; that is to say, the site mean directions are located on the great circle. This is typical of both the Kiya and Bateni fragments of the Kuznetsk Alatau island arc system (Fig. 8) and suggests that, in the Cambrian, they experienced a movement which can be described as a simple rotation around the same Euler pole (strike-slip). The virtual geomagnetic poles for the Bateni terrane form a trend reflecting this regularity. In the Cambrian, the northern paleomagnetic pole wandered southeastward from the water area of Madagascar Island to the southeastern coast of Australia (Fig. 8). It is well-known that any version of the Phanerozoic APWP of the Siberian craton starts here (Cocks and Torsvik, 2007; Pechersky and Didenko, 1995; Smethurst et al., 1998).

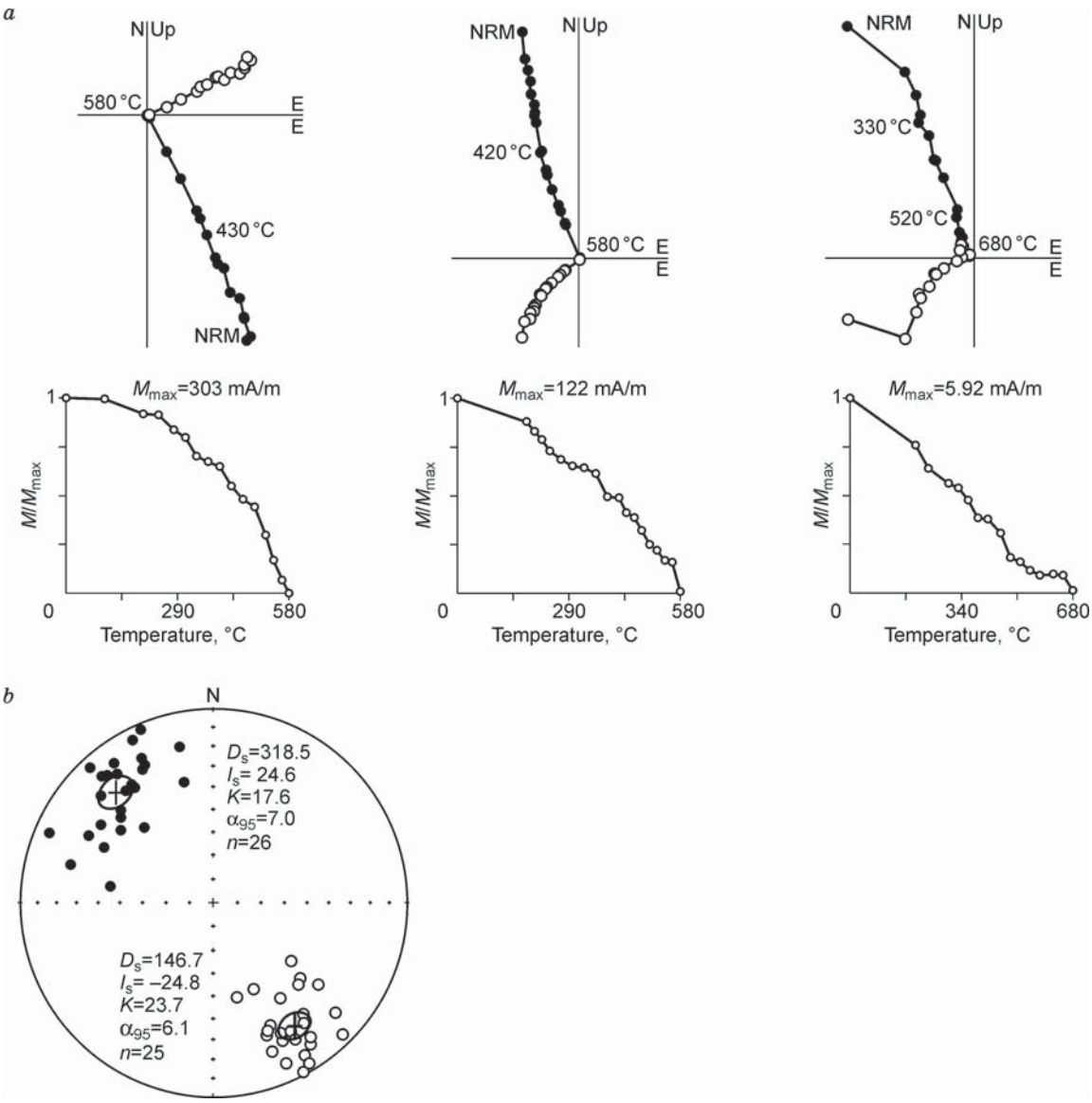


Fig. 7. Typical orthogonal diagrams and associated NRM(*T*) decay plots illustrating stepwise thermal demagnetization of the sandstones and siltstones from the Bateni Formation (*a*); stereoplots (*b*) show the distribution of the unit vectors of component A3 from the Bateni Formation and the tuffaceous rocks of the Bograd and Beibuluk localities (illustration of the reversal test).

The APWP of the Bateni terrane, constructed by the spline method (Torsvik and Smethurst, 1999), differs from that of the Kiya terrane only in the details (Fig. 9). The considerable differences at the ends of the paths are due to the smoothing procedure (Torsvik and Smethurst, 1999) rather than real tectonics. The APWP permit calculating and comparing the main kinematic parameters of the horizontal movements of the blocks on the eastern slope of the Kuznetsk Alatau (Fig. 9). From the Late Vendian to the Early Ordovician, the fragments of the Kuznetsk Alatau island arc, located near the Equator, drifted from the Northern to the Southern Hemisphere. Considering the age of the components, the velocity of this drift might have averaged around 5–6 cm/yr, peaking in the Early–Middle Cambrian (up to 15 cm/yr). During the same time, the angular component of the velocity reaches a maximum: somewhat more than 2 deg/myr for the Kiya fragment and up to 6 deg/myr for the Bateni fragment (Fig. 9). In general, the southward drift of the island arc was accompanied by a clockwise rotation with a possible amplitude of more than 50°.

The kinematics described agrees with the paleomagnetic data on the Tersa and Zolotoi Kitat fragments of the Kuznetsk Alatau paleoisland arc (Table 2). The virtual geomagnetic poles for the Vendian–Cambrian and Middle–Late Cambrian ophiolites of the Middle Tersa pluton are very much the same as the corresponding paleomagnetic poles for the Bateni and Kiya terranes (Fig. 8). Thus, the terranes on the eastern slope of the Kuznetsk Alatau were, undoubtedly, fragments of one island arc system, which deformed in one style while experiencing a predominantly transform (strike-slip) movement. The slight angular difference in the positions of the Vendian–Cambrian paleomagnetic poles suggests the regional scale of this

Table 2. Terranes of the Kuznetsk Alatau paleoisland arc: mean paleomagnetic directions, paleomagnetic poles, and paleolatitudes

Formation	Age, Ma	*N*	Paleomagnetic direction			Paleomagnetic pole			PL	References
			D	*I*	α_{95}	Plat	Plong	A_{95}		
Zolotoi Kitat terrane										
Edinis	540	4	64.1	–28.6	6.1	–1.1	26.8	5.3	15.3 ± 5.3	Kazansky, 2002
Bol'shoi Kitat, Chernoosipovskaya, Kitat	500	17	98.2	–23.2	3.5	–14.6	177.2	3.2	–12.1 ± 3.2	Kazansky, 2002
Kiya terrane										
Ust'–Anzas, Chumai	540	7	23.6	–16.7	4.9	–23.0	62.5	4.1	8.5 ± 4.1	Kazansky et al., 2003a
Lower part of the Berikul' Formation	510	2	181.5	–1.1	4.5	–35.0	86.2	3.2	–0.5 ± 3.2	Kazansky et al., 2003a
Upper part of the Berikul' Formation	505	1	164.5	–19.8	7.1	–43.1	109.1	5.4	–10.2 ± 5.4	Kazansky et al., 2003a
Smirnovskaya	480	8	147.5	–29.7	5.5	–43.3	133.5	4.7	–16.0 ± 4.7	Kazansky et al., 2003a
Tersa terrane										
Middle Tersa dikes	540	2	222.5	20.3	10.5	–15.6	44.6	8.0	10.5 ± 8.0	Kazansky et al., 2003b
Middle Tersa dikes	500	3	145.5	–19.0	8.4	–37.4	132.8	7.2	–9.8 ± 7.2	Kazansky et al., 2003b
Bateni terrane										
Kolodzhul, Shipilinsk	520	9	10.7	–13.7	8.6	–27.9	77.4	7.9	7.1 ± 7.9	This paper
Sladkie Koren'ya, Bezymyannaya	505	7	351.9	12.0	4.3	–41.2	100.6	3.7	–6.2 ± 3.7	This paper
Bateni, and undifferentiated turbidite strata	500	6	319.6	24.5	9.9	–37.9	144.1	8.9	–13.0 ± 8.9	This paper; Metelkin et al., 2000

Note. "Age" is the presumed age of the paleomagnetic pole; *N*, number of sampling sites (virtual geomagnetic poles); the paleomagnetic direction is paleonorthward in the given polarity mode: *D*, declination; *I*, inclination; α_{95}, radius of the 95% cone of confidence; paleomagnetic pole shows the northern paleopole (averaged from the corresponding virtual geomagnetic poles; at $N < 3$ the pole was calculated as a virtual geomagnetic pole for the mean direction): Plat, latitude; Plong, longitude; A_{95}, radius of the 95% cone of confidence; PL, terrane paleolatitude at the chosen polarity of the paleomagnetic directions (positive, Northern Hemisphere; negative, Southern Hemisphere).

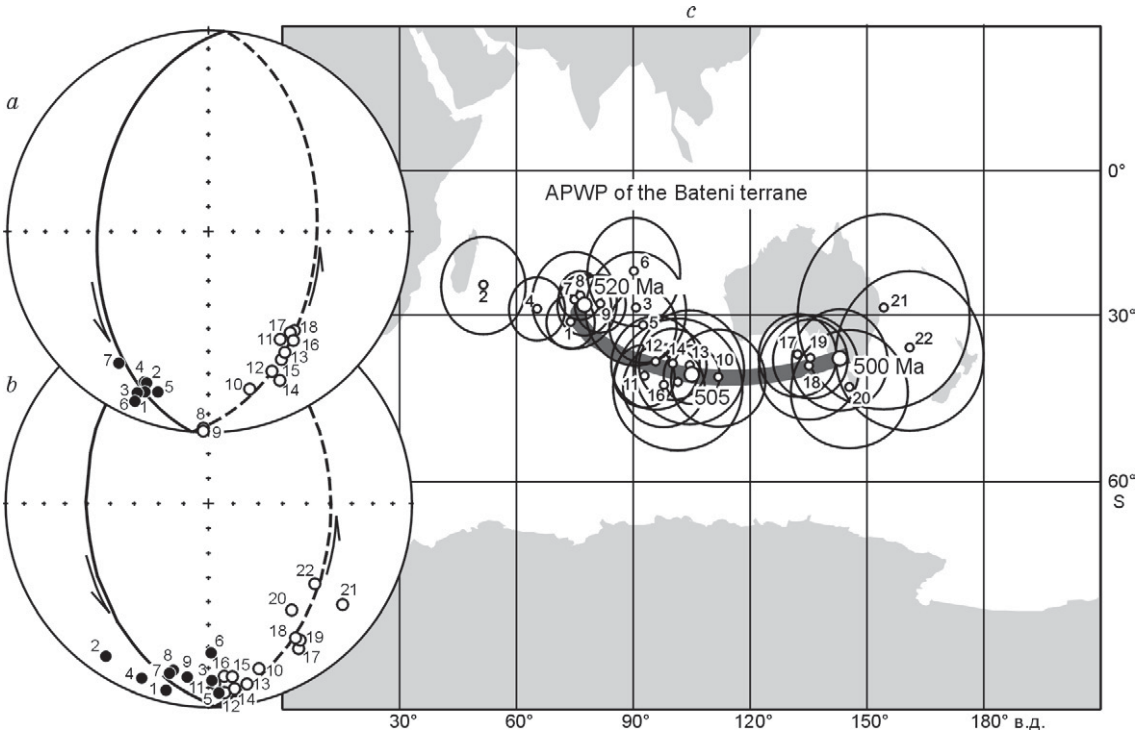


Fig. 8. Variation trends for the paleomagnetic directions of the Kiya block, after (Kazansky et al., 2003a) (*a*) and Bateni block (*b*) (numbers on the stereoplots correspond to the succession in Table 1); trend of the virtual geomagnetic poles (pole numbers correspond to those of the directions on the stereoplot) of the Bateni terrane (*c*).

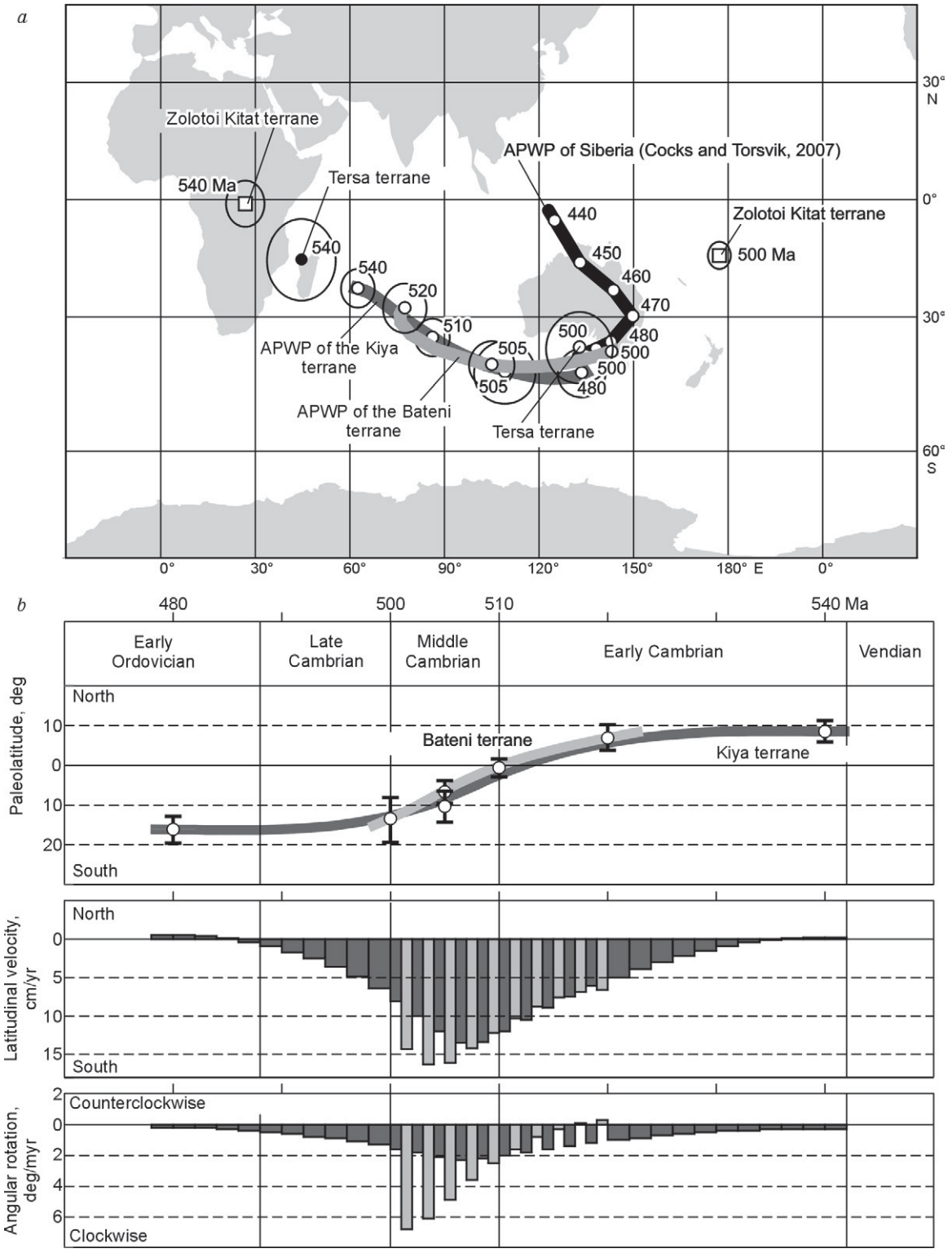


Fig. 9. Paleotectonic analysis of the paleomagnetic data on the Late Vendian–Early Ordovician of the Kuznetsk Alatau: *a*, comparison between the Vendian–Early Paleozoic APWP intervals for the Siberian craton (Cocks and Torsvik, 2007), Kiya terrane (Kazansky et al., 2003a), and Bateni terrane, as well as the positions of the paleomagnetic poles for the Zolotoi Kitat (Kazansky, 2002) and Tersa blocks (Kazansky et al., 2003b); *b*, kinematic parameters of the drift of the Kiya (dark) and Bateni (light) terranes in the early Paleozoic, calculated from APWP: paleolatitudinal (top diagram) drift history, latitudinal velocity (middle diagram), and angular (lower diagram) velocity.

process, that is to say, the relatively small strike-slip movement of the island arc fragments with respect to one another, as presumed in (Alabin and Kalinin, 1999; Kazansky, 2002). The poles of the latest, Cambrian–Ordovician, stage in the development of the system are within an isometric and compact area, which coincides with that containing the Cambrian–Ordovician paleomagnetic poles of Siberia (Fig. 8). This agrees with the Early Caledonian accretion of the arc to the craton and the formation of a large orogenic structure in southwestern Siberia (geographic coordinates) (Dobretsov and Buslov, 2007; Kungurtsev et al., 2001; Vladimirov et al., 2003).

The Vendian–Cambrian (Edinis Formation) paleomagnetic pole for the Zolotoi Kitat terrane on the western slope of the Kuznetsk Alatau is located on the same great circle as the poles in the eastern part of the orogen and occupies the extreme position (Fig. 9). Therefore, the kinematics of its relative drift can also be considered in the light of the general strike-slip model. Note that the Middle–Late Cambrian (510–490 Ma) poles along the Bol'shoi Kitat, Chernoosipovskaya, and Kitat Formations of the Zolotoi Kitat terrane shift by a considerably larger angle with respect to the coeval poles on the eastern slope of the Kuznetsk Alatau and are located aside from the APWP of Siberia (Fig. 8). The simplest explanation for this difference is the relatively younger, Late Paleozoic–Mesozoic, strike-slip activity along the Kuznetsk–Altai fault. It determined the difference in the positions of the Early Paleozoic paleomagnetic poles for the structures located to its different sides.

Thus, the present structure of the Kuznetsk Alatau is related to strike-slip activity, which was the most intense during the development of the island arc in the Cambrian and its accretion in the early Ordovician.

Conclusions

1. The restored APWP of the Bateni fragment (back-arc) is very much the same as that for the Kiya fragment (paleoarc center), suggesting that the tectonics of the Kuznetsk Alatau island arc developed in one style in the Cambrian. During this time the island arc system moved from ~10° N to ~10° S with a simultaneous clockwise rotation by more than 50°.

2. The regularities in the position of the paleomagnetic poles for the Kuznetsk Alatau show that the drift of the Cambrian island arc fragments, like that of the entire paleoisland arc, resulted from strike-slip activity. This strike-slip system at the Cambrian stage of the active island arc with specific volcanism might have formed owing to the predominantly oblique subduction and subsequent accretion. This regime has been discussed in many publications (Kungurtsev et al., 2001; Metelkin et al., 2009; Vladimirov et al., 2003).

3. The general structure of the Kuznetsk Alatau had formed by as early as the Cambrian–Ordovician. This is evidenced by the similarity of the Late Cambrian paleomagnetic poles for the main structures on the eastern slope of the Kuznetsk Alatau to one another and the Siberian poles. However, a difference is observed in the positions of the Late Cambrian–Ordovician poles on the eastern (Kiya, Tersa, Bateni terranes) and western (Zolotoi Kitat terrane) sides of the Kuznetsk Alatau; it might be due to a younger strike-slip. This one is probably related to the reactivation of the main fault in the Kuznetsk Alatau (Kuznetsk–Altai deep fault) in the late Paleozoic or in the Mesozoic. The causes of the strike-slip tectonics which affected the region in the late Paleozoic, at the Hercynian collisional stage, were discussed in (Buslov et al., 2003; Kazansky, 2002; Vladimirov et al., 2003): it was attributed to the clockwise rotation of the Siberian Plate. The paleomagnetic data also show that the Paleozoic deformation mechanism of the structures confined to the strike-slips in the western Altai–Sayan region, up to the Kuznetsk–Altai fault, persisted into the early Mesozoic (Kazansky et al., 2005; Metelkin et al., 2009) and even the Jurassic–Cretaceous (Metelkin et al., 2010).

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